CODEFLOW PSUT · HANDOUT 03

# Library System UI/UX Wireframes & Redesign Exercise

Industry Practices Using the SDLC · UI/UX Design · November 4, 2026

PART A
**The current design**
Student-facing flawed screens

PART B
**Your turn**
Analysis & sketching activity

PART C
**Instructor reveal**
Reference redesign — after the activity

PART D
**Answer key**
Comparison & principles

Fictional university library system created for teaching. Sample books are real titles used as placeholder data.

READ FIRST

# How to use this pack

## Teaching sequence

A **Study the current design (pages 3–8).** Seven screens of a library app that technically meets the requirements. Walk through them in order as Omar, the student below.

B **Analyse and redesign in pairs (pages 9–11).** List problems, explain their impact, prioritise, sketch a better flow. 15–20 minutes.

C **Instructor reveal (page 12 onward).** A reference redesign, then the answer key (Part D). Do not show these pages before the activity.

**Instructor tip:** print or share pages 1–11 for students. Keep pages 12–23 for projection after the activity.

## The scenario

It is **Wednesday, 4 November 2026**. **Omar Khalid** (student ID 20230417) needs *Introduction to Algorithms* for an assignment. He opens the library app on his phone between lectures. He already has some reservations and loans. Later, a **librarian** processes the day's requests at the circulation desk.

Requirement IDs (e.g. `FR-07`) refer to Handout 02 — the SRS.

## Sample data used in both designs

| Book | Copies | State on 4 Nov |
|---|---|---|
| **Introduction to Algorithms** (4th ed.)<br>Cormen, Leiserson, Rivest, Stein · CS | 3 | 2 available, 1 on loan |
| **Clean Code**<br>Robert C. Martin · CS | 2 | 0 available · Omar is #2 on waitlist |
| **Software Engineering** (10th ed.)<br>Ian Sommerville · CS | 4 | Omar's copy is ready for pickup |
| **Database System Concepts** (7th ed.)<br>Silberschatz, Korth, Sudarshan · CS | 2 | Omar's loan, due 7 Nov 2026 |
| **Discrete Mathematics and Its Applications**<br>Kenneth H. Rosen · Mathematics | 3 | Omar's loan, due 2 Nov — overdue |
| **Calculus: Early Transcendentals**<br>James Stewart · Mathematics | 3 | Returned by Omar on 6 Oct |

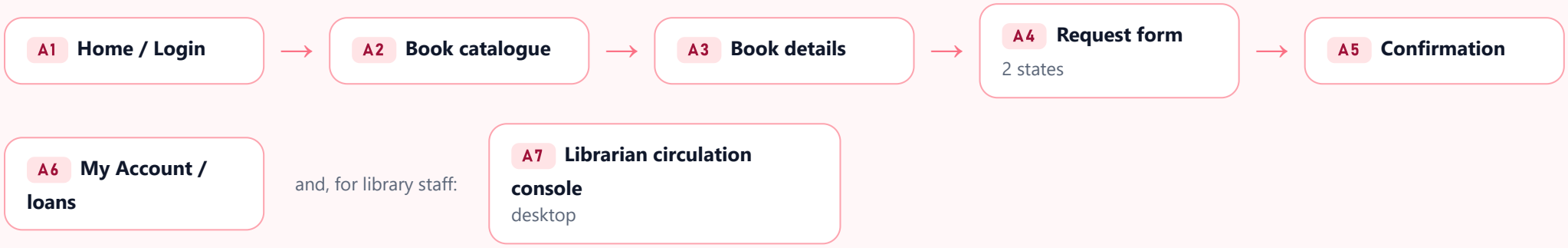
**Illustrative policy values** (from SRS business rules, not real PSUT rules): 14-day loans · reservations held 48 h after "ready" · example fine rate 0.25 JOD per overdue day.

PART A

# The current design

Student-facing · 7 screens

This is the library app as it exists today. It was built from the requirements by a team that focused on functionality. **Every screen works.** Your job is to decide whether it is *usable* — and why or why not.




### Walk through it as Omar

Goal: find out if *Introduction to Algorithms* is available and reserve it.

### Then check his account

Goal: know what to collect, what is due and whether he owes anything.

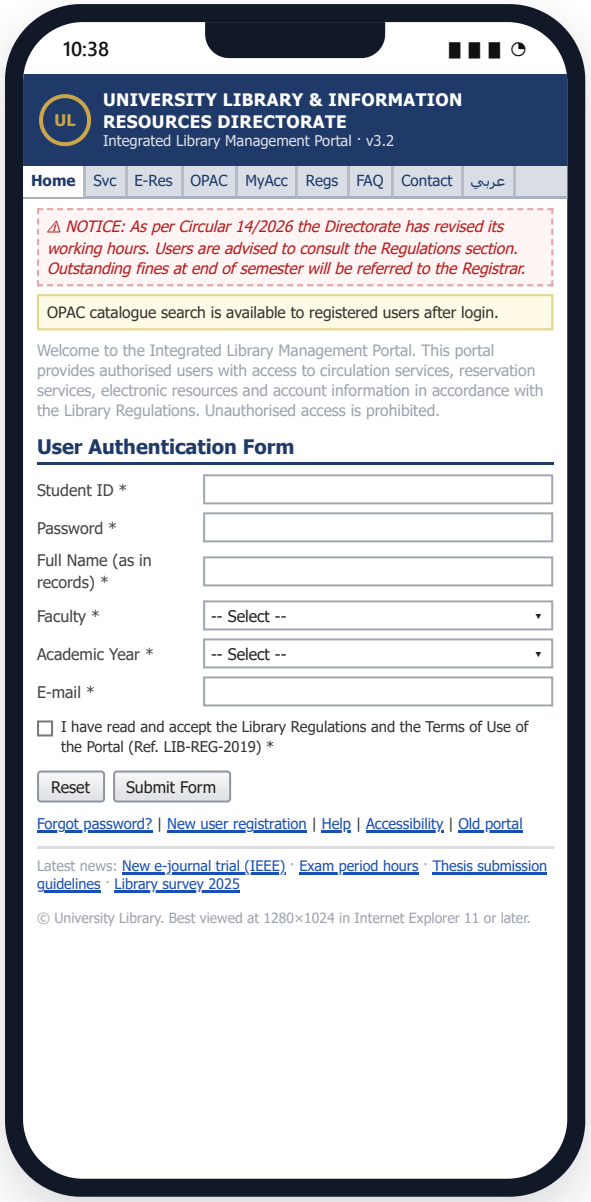
### Then be the librarian

Goal: prepare today's requests and hand books over quickly and correctly.

PART A

# Home / Login → Catalogue

Omar opens the app to look for a book




A1

## Home / Login

First screen when the app opens

→ after Submit Form




A2

## Book catalogue

Search results for “algorithms”

Your notes

PART A

# Book details → Request form

Omar opens the first result and tries to reserve it

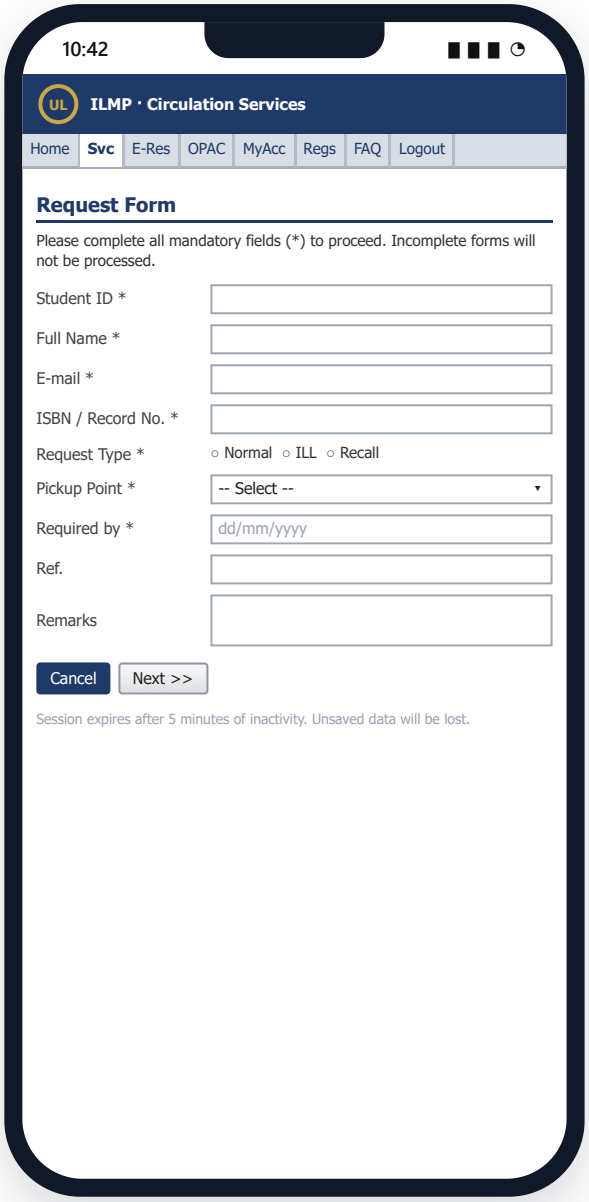



A3

**Book details**

Introduction to Algorithms

after Place Request




A4

**Request form**

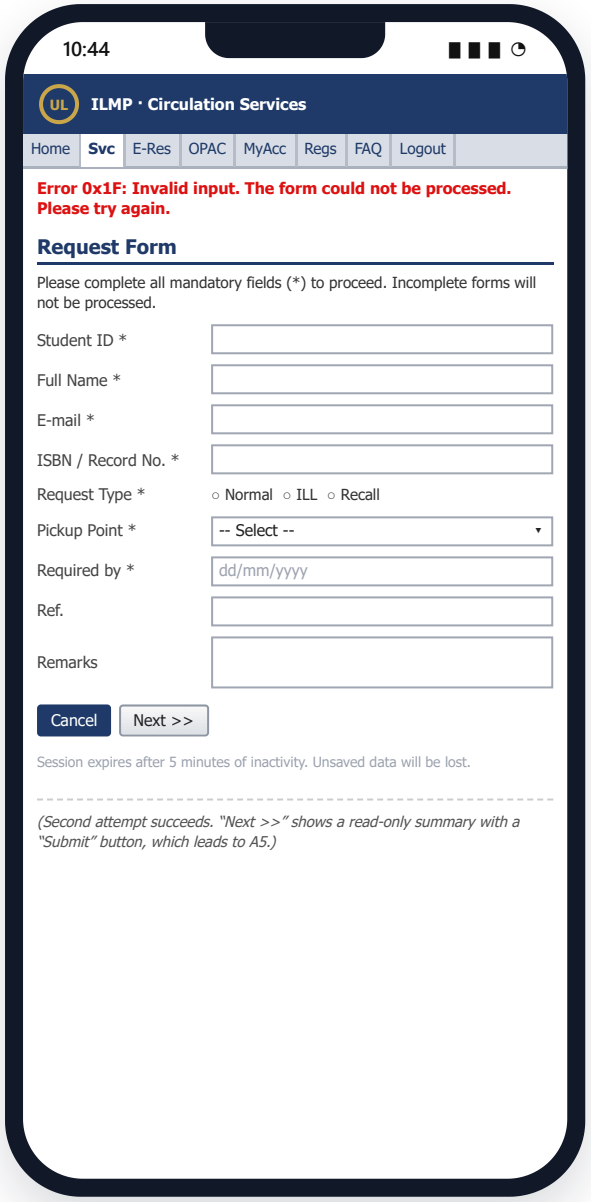
Opened from “Place Request”

Your notes

PART A

# Request form (after Next) → Confirmation

Omar typed the date as 6-11-2026

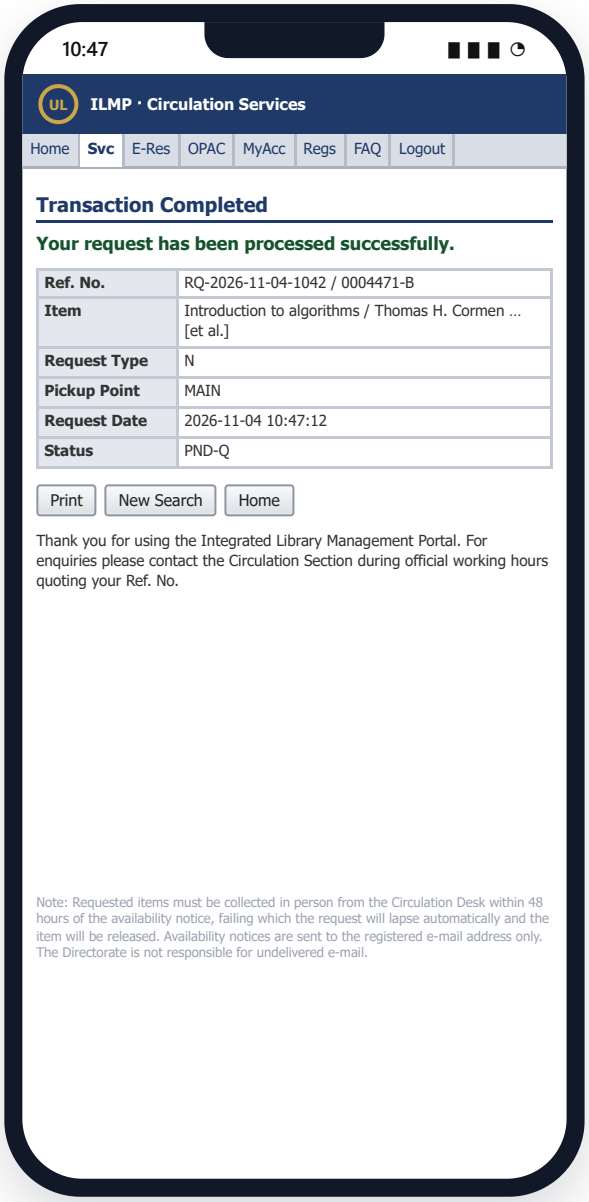



A4

**Request form — after “Next”**

Same screen, returned with an error

→ after Submit




A5

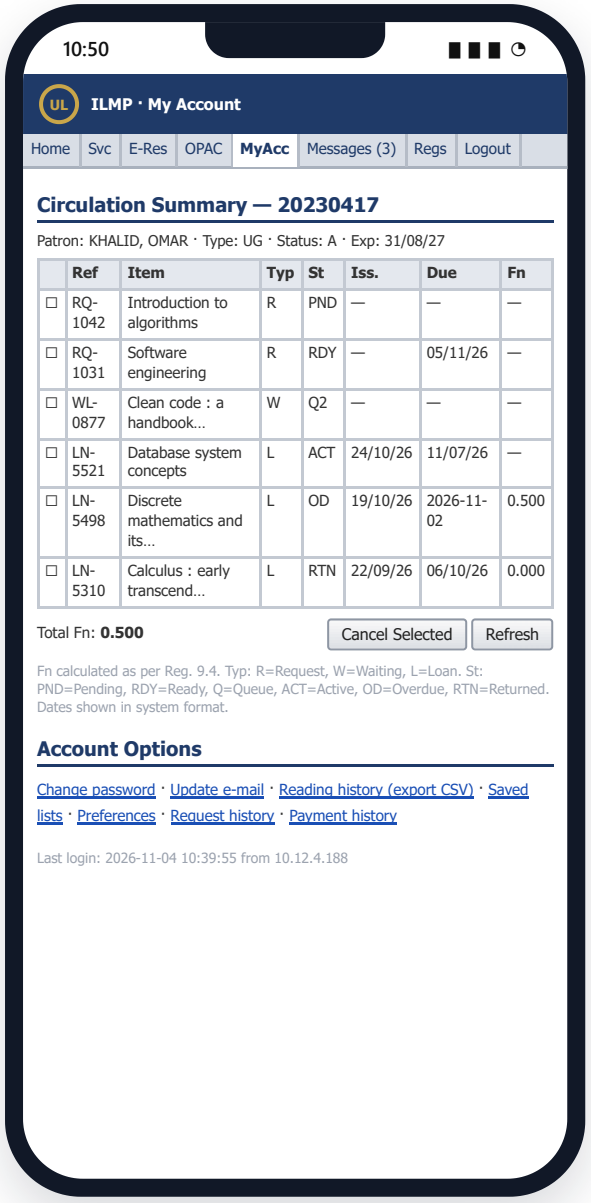
**Confirmation**

Shown after the request is submitted

Your notes

# My Account / loan tracking

Omar checks what he needs to collect, return or pay




A6

**My Account**

Reached via “MyAcc” in the menu

Your notes

## Try these tasks as Omar

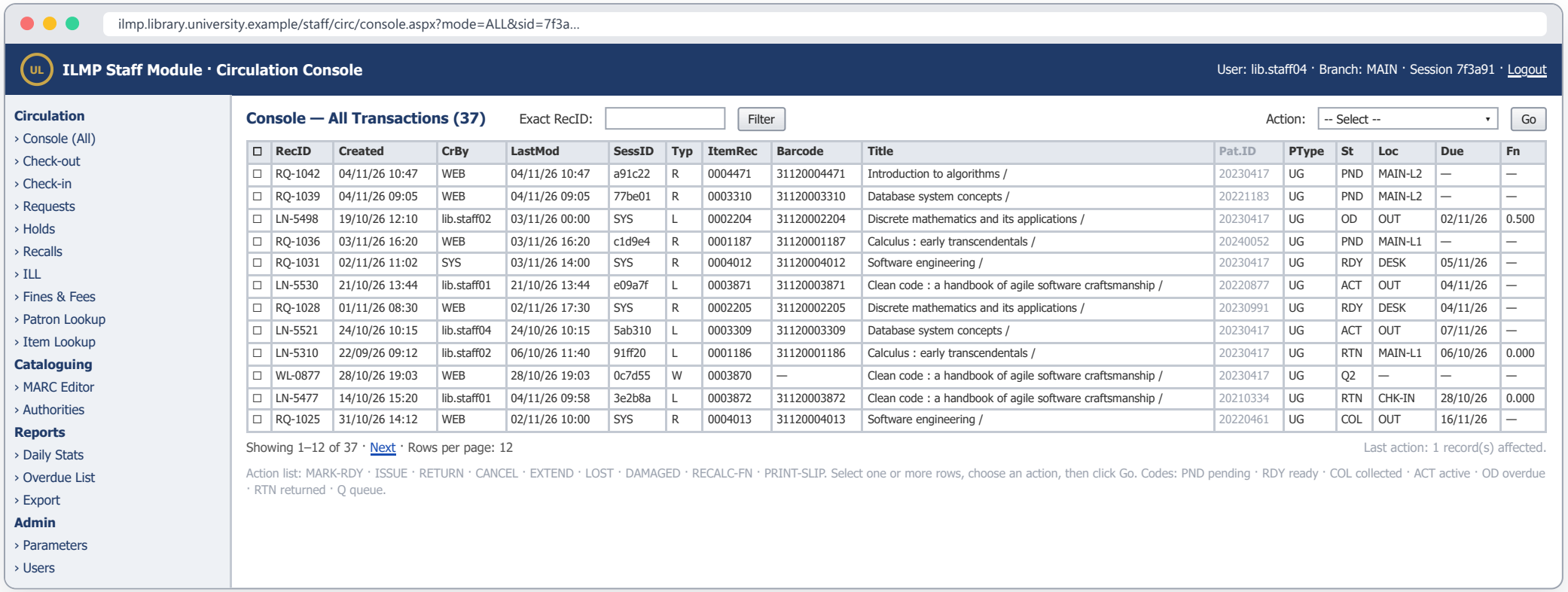
1. Which book can you collect **today**, and until when?
2. When is *Database System Concepts* due?
3. Do you owe a fine? How much, in what currency, and why?
4. Where is *Introduction to Algorithms* in the process?
5. You have 3 messages. What do they say?

Time yourself. Note every moment you had to guess.

PART A

# Librarian circulation console

Desktop · used at the circulation desk




| ☐ | RecID | Created | CrBy | LastMod | SessID | Typ | ItemRec | Barcode | Title | Pat.ID | PType | St | Loc | Due | Fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ☐ | RQ-1042 | 04/11/26 10:47 | WEB | 04/11/26 10:47 | a91c22 | R | 0004471 | 31120004471 | Introduction to algorithms / | 20230417 | UG | PND | MAIN-L2 | — | — |
| ☐ | RQ-1039 | 04/11/26 09:05 | WEB | 04/11/26 09:05 | 77be01 | R | 0003310 | 31120003310 | Database system concepts / | 20221183 | UG | PND | MAIN-L2 | — | — |
| ☐ | LN-5498 | 19/10/26 12:10 | lib.staff02 | 03/11/26 00:00 | SYS | L | 0002204 | 31120002204 | Discrete mathematics and its applications / | 20230417 | UG | OD | OUT | 02/11/26 | 0.500 |
| ☐ | RQ-1036 | 03/11/26 16:20 | WEB | 03/11/26 16:20 | c1d9e4 | R | 0001187 | 31120001187 | Calculus : early transcendentals / | 20240052 | UG | PND | MAIN-L1 | — | — |
| ☐ | RQ-1031 | 02/11/26 11:02 | SYS | 03/11/26 14:00 | SYS | R | 0004012 | 31120004012 | Software engineering / | 20230417 | UG | RDY | DESK | 05/11/26 | — |
| ☐ | LN-5530 | 21/10/26 13:44 | lib.staff01 | 21/10/26 13:44 | e09a7f | L | 0003871 | 31120003871 | Clean code : a handbook of agile software craftsmanship / | 20220877 | UG | ACT | OUT | 04/11/26 | — |
| ☐ | RQ-1028 | 01/11/26 08:30 | WEB | 02/11/26 17:30 | SYS | R | 0002205 | 31120002205 | Discrete mathematics and its applications / | 20230991 | UG | RDY | DESK | 04/11/26 | — |
| ☐ | LN-5521 | 24/10/26 10:15 | lib.staff04 | 24/10/26 10:15 | 5ab310 | L | 0003309 | 31120003309 | Database system concepts / | 20230417 | UG | ACT | OUT | 07/11/26 | — |
| ☐ | LN-5310 | 22/09/26 09:12 | lib.staff02 | 06/10/26 11:40 | 91ff20 | L | 0001186 | 31120001186 | Calculus : early transcendentals / | 20230417 | UG | RTN | MAIN-L1 | 06/10/26 | 0.000 |
| ☐ | WL-0877 | 28/10/26 19:03 | WEB | 28/10/26 19:03 | 0c7d55 | W | 0003870 | — | Clean code : a handbook of agile software craftsmanship / | 20230417 | UG | Q2 | — | — | — |
| ☐ | LN-5477 | 14/10/26 15:20 | lib.staff01 | 04/11/26 09:58 | 3e2b8a | L | 0003872 | 31120003872 | Clean code : a handbook of agile software craftsmanship / | 20210334 | UG | RTN | CHK-IN | 28/10/26 | 0.000 |
| ☐ | RQ-1025 | 31/10/26 14:12 | WEB | 02/11/26 10:00 | SYS | R | 0004013 | 31120004013 | Software engineering / | 20220461 | UG | COL | OUT | 16/11/26 | — |

A7

## Librarian circulation console

Omar is at the desk to collect his book. The librarian must find his ready item and issue it.

Your notes

PART B

# YOUR TURN: FIX THE EXPERIENCE

Work in pairs · 15–20 minutes

## Instructions

1. Examine the flawed wireframes (A1–A7).
2. Identify at least **five** usability problems.
3. Explain **why** each problem affects the user.
4. **Prioritise** the most serious problems.
5. Sketch a redesigned **homepage, catalogue or reservation flow**.
6. Explain the reasoning behind your decisions — link them to user goals or requirements in Handout 02.

**Remember:** a problem is not "I don't like the colour". Describe what the user is trying to do, what goes wrong, and what it costs them (time, errors, confusion, exclusion).

## Discussion questions

- What is the user's primary goal?
- Which steps are unnecessary?
- Where should authentication happen?
- How should the interface display availability?
- How should unavailable books be handled?
- How should physical collection be communicated?
- How can the student monitor their borrowing period?
- How should the system communicate errors?
- What should change on mobile?
- How can the librarian complete routine actions faster?

PART B

# Problem log

Fill in at least five rows, then rank them

| Rank | Screen | What is the problem? | Who is affected, and how? (time, errors, confusion, exclusion) | Requirement / principle | Your proposed fix |
|---|---|---|---|---|---|
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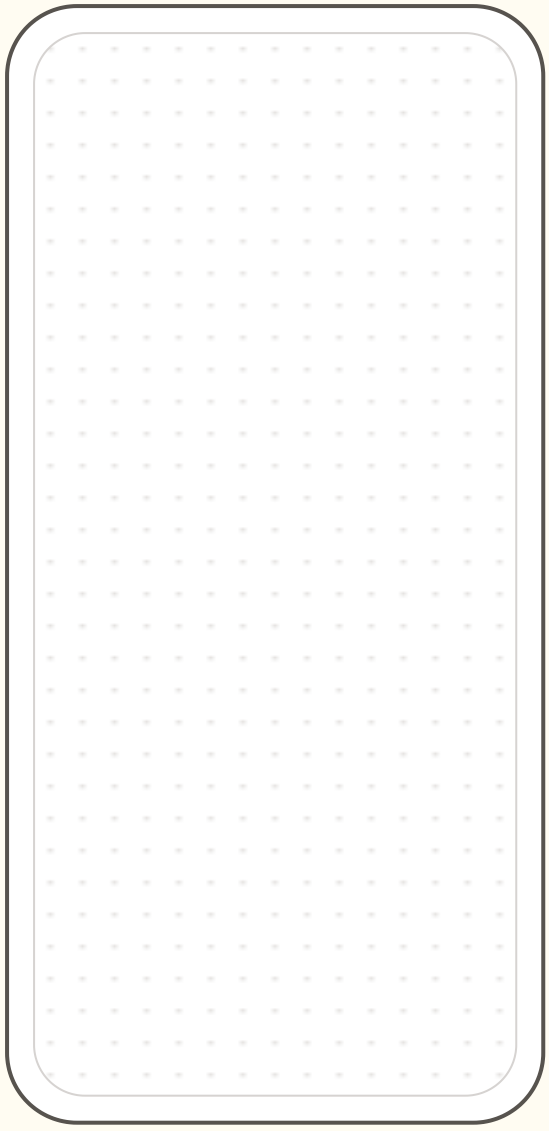
**Severity guide. High** — blocks the task or causes a wrong outcome. **Medium** — slows the user or causes avoidable errors. **Low** — cosmetic or minor annoyance.

**Is it bad UX or missing functionality?** If the feature exists but is hard to find or use, it is a UX problem. If the system cannot do it at all, it is a functional gap. If it contradicts the SRS, it is a requirements problem.

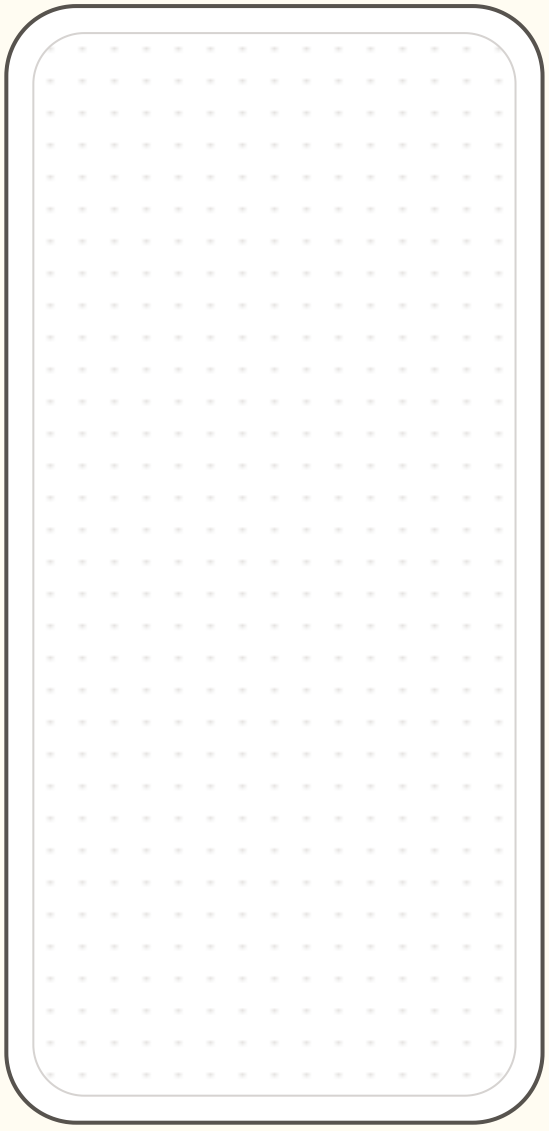
PART B

# Sketch canvas

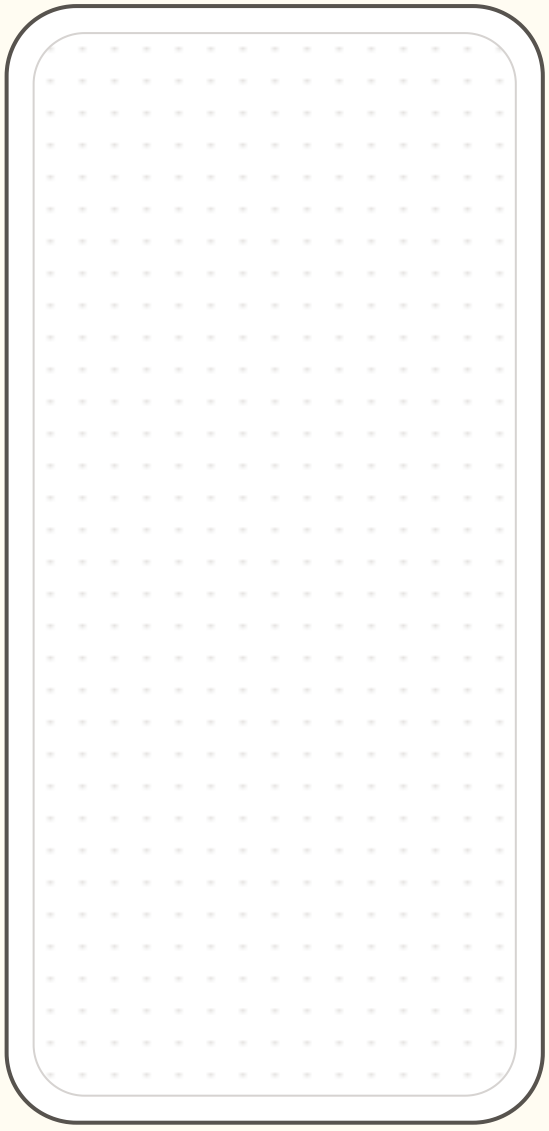
Low-fidelity: boxes, lines and labels. No colours needed.



Screen 1: _____



Screen 2: _____



Screen 3: _____




# STOP — Instructor Reveal

Do Not Continue Until After the Redesign Activity.

Parts C and D contain the reference redesign and the answer key.

PART C

# The improved design — reference solution

Mobile-first student app · desktop librarian dashboard

The redesign keeps **exactly the same capabilities** as Part A — search, details, reservation, waitlist, notifications, loans, fines and librarian processing. What changes is **order, priority, wording and feedback**. It is a proposed reference, not the only correct answer.

## Core student flow — authenticate only when it is needed

C1 **Explore** — no sign-in → C2–C3 **Select book** — see real copies → C3 **Reserve book** — primary action → C4 **Authenticate** — university sign-in → C4 **Back to reservation** — same book kept — FR-08 → C5 **Confirm** — status + next steps

### 1 · Goal first

The first screen answers “do you have this book?” — search is the hero, sign-in is optional.

### 2 · Honest status

Availability counts real copies. Reservation states separate “Preparing” from “Ready for pickup”.

### 3 · Physical world visible

Pickup location, hours, 48-hour window and ID requirement appear wherever a decision is made.

### 4 · One primary action

Each screen has one clear, labelled main button. Destructive actions ask for confirmation.

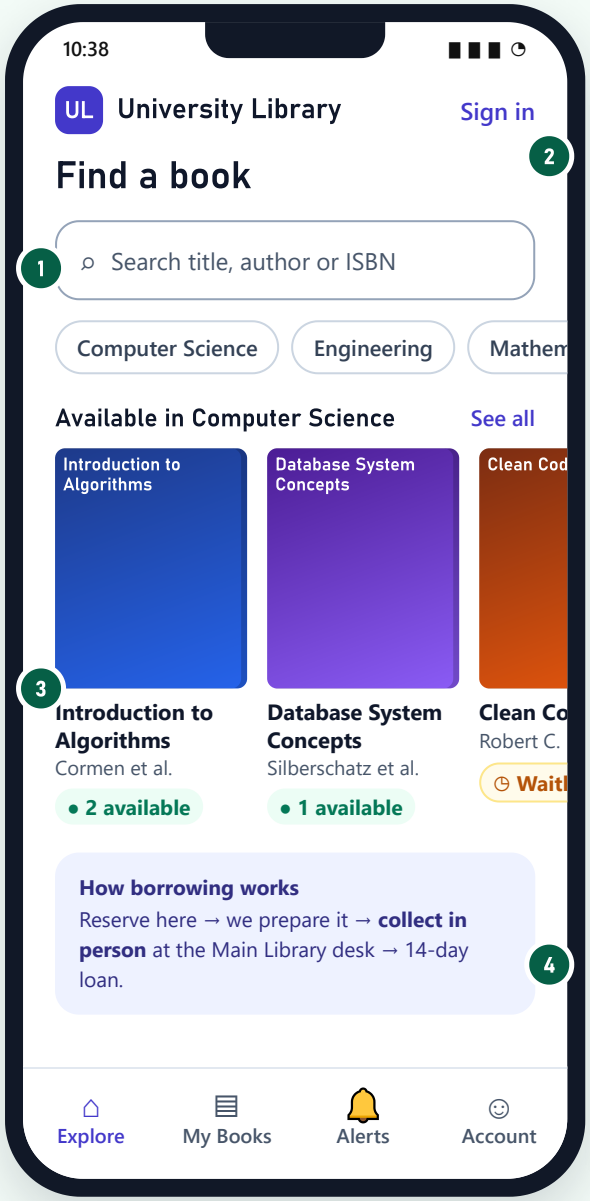
**Screens in this part:** C1 Explore · C2 Search results (+ empty state) · C3 Book details (available / unavailable) · C4 Contextual sign-in · C5 Confirmation · C6 My Books (+ cancel dialog) · C7 Notifications (+ loading & error states) · C8 Librarian dashboard · Visual system.

Green numbered markers (1) point to design decisions explained beside each screen.

PART C

# Explore → Search results

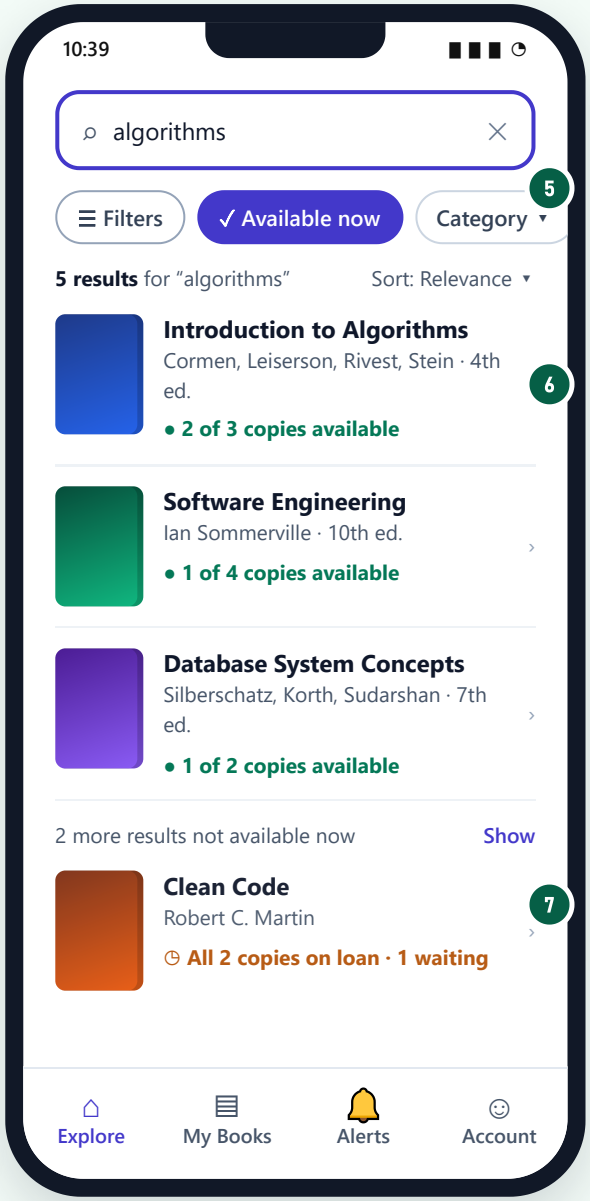
Guest can browse and search without signing in FR-01 FR-07




C1

## Explore (guest)

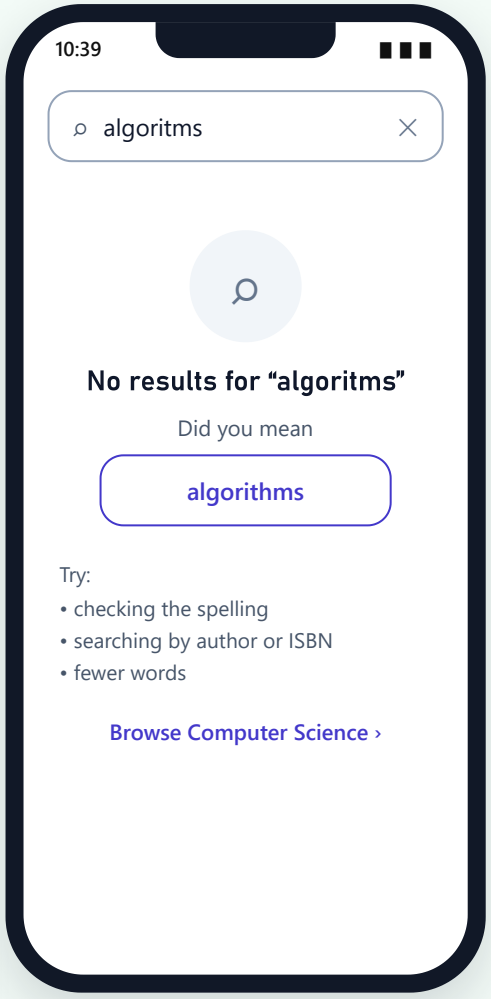
Search is the hero; no login wall




C2

## Search results

"Available now" filter on




C2

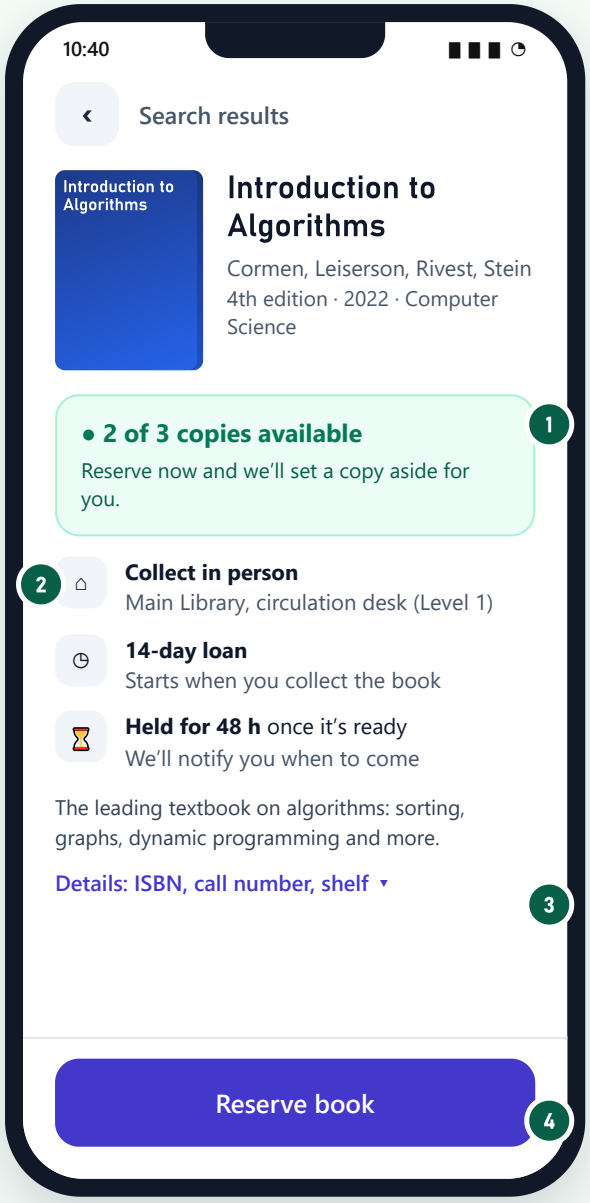
## Empty state

Helps the user recover

1. **Search first.** The user's main goal is the largest target on screen.
2. **Sign-in is optional** here; it appears only when needed FR-07 .
3. **Availability on every card** — text + icon, not colour alone.
4. **Expectations early:** pickup is physical; tab bar has labels.
5. **Filters visible** as chips; advanced options behind one button.
6. **Real copy counts** ("2 of 3") instead of "In/Out" FR-04 .
7. **Unavailable** books are still shown, with the waitlist explained.

PART C

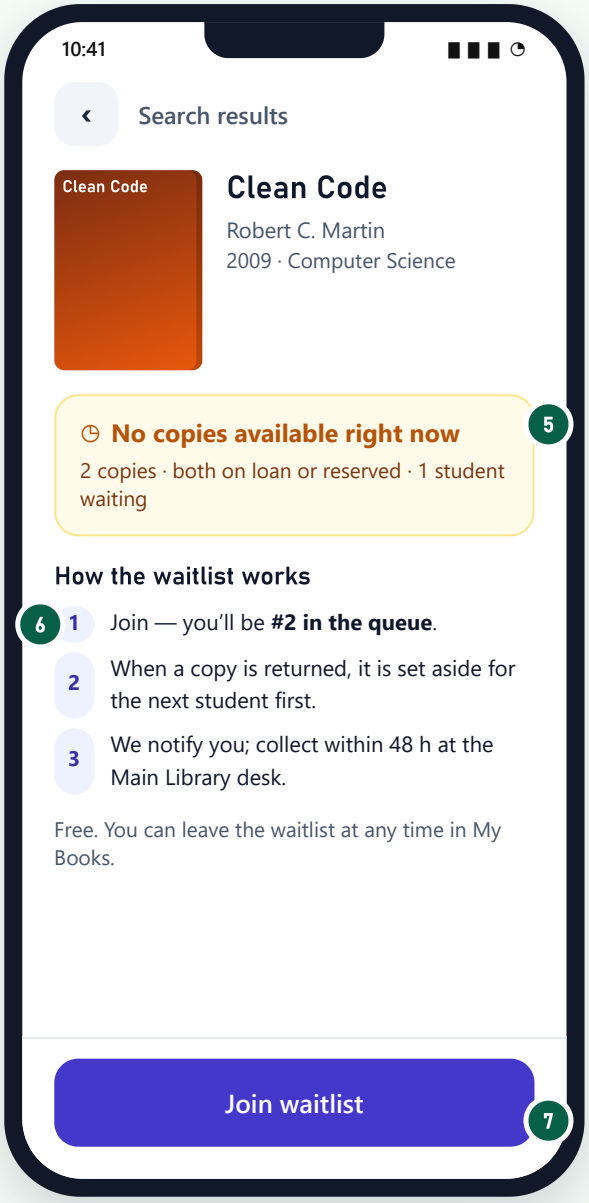
# Book details — available vs. unavailable




C3

**Available**

Introduction to Algorithms




C3

**Unavailable**

Clean Code

One clear primary action, with defined behaviour FR-03 FR-09 FR-11

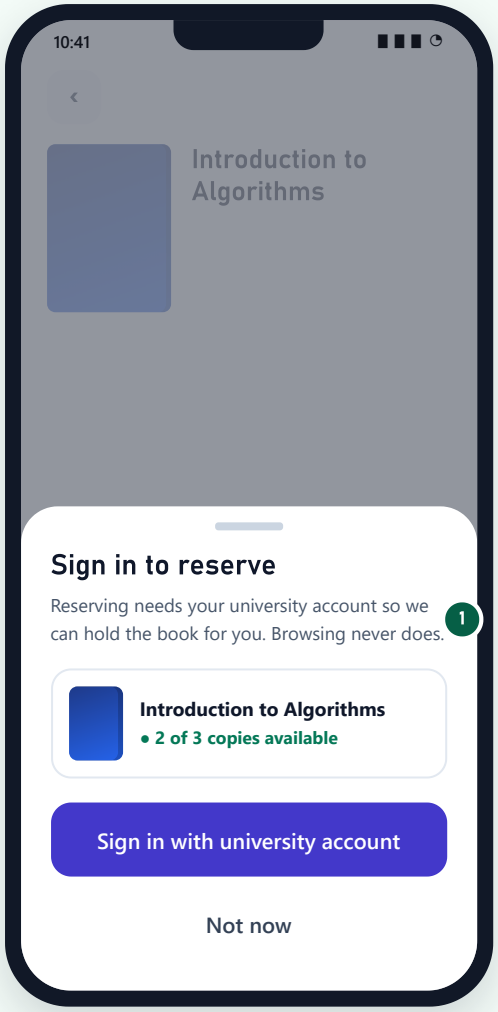
1. **Availability first,** as copies — answers "can I get it?" instantly.
2. **Physical pickup, loan length and hold window** are visible before the user commits BR-01 BR-02 BR-05.
3. **Bibliographic detail is collapsed** — available, but not competing with the task.
4. **Sticky primary button** in the thumb zone; label says exactly what happens.
5. **Unavailable is explained** (why, how many waiting) instead of just "Out".
6. **Waitlist behaviour is defined** in plain steps, matching BR-07 FR-15.
7. **Same position, different action:** "Join waitlist" replaces "Reserve book" — never both.

If the last copy is taken by someone else while Omar is reading, tapping **Reserve book** shows "This copy was just reserved" and offers the waitlist FR-10.

PART C

# Contextual authentication

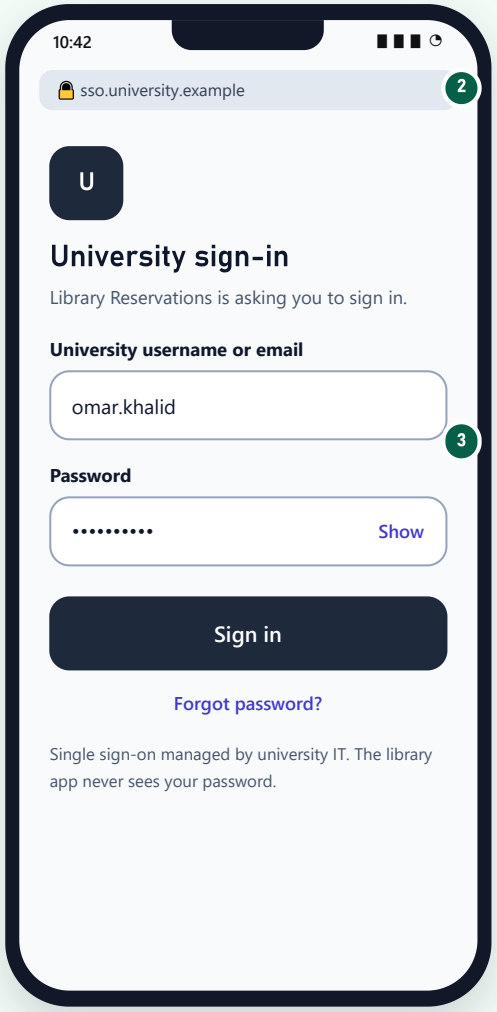
Sign in only when reserving — and come back to the same book FR-05 FR-08




C4·a

**Tap “Reserve book”**

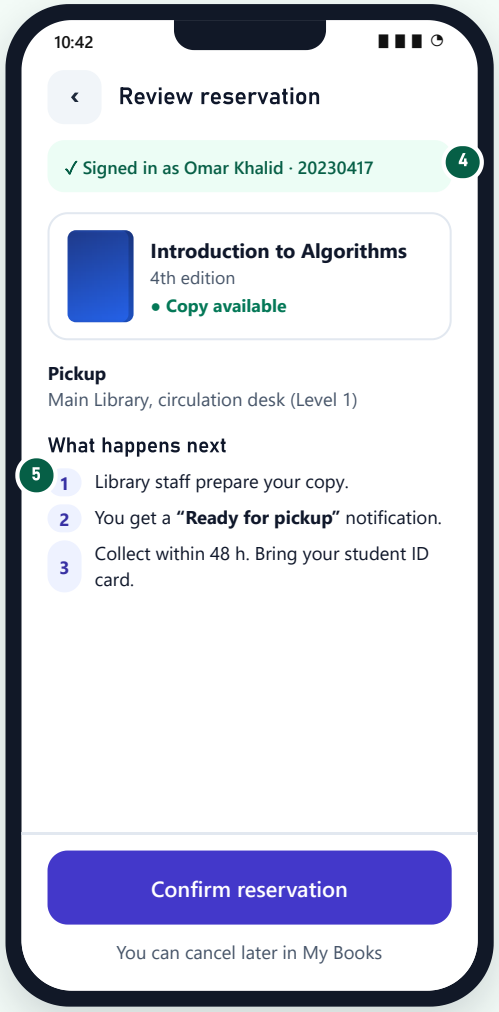
Not signed in → sheet




C4·b

**Approved university login**

Generic SSO page (illustrative)




C4·c

**Back in context**

Same book, one confirm tap

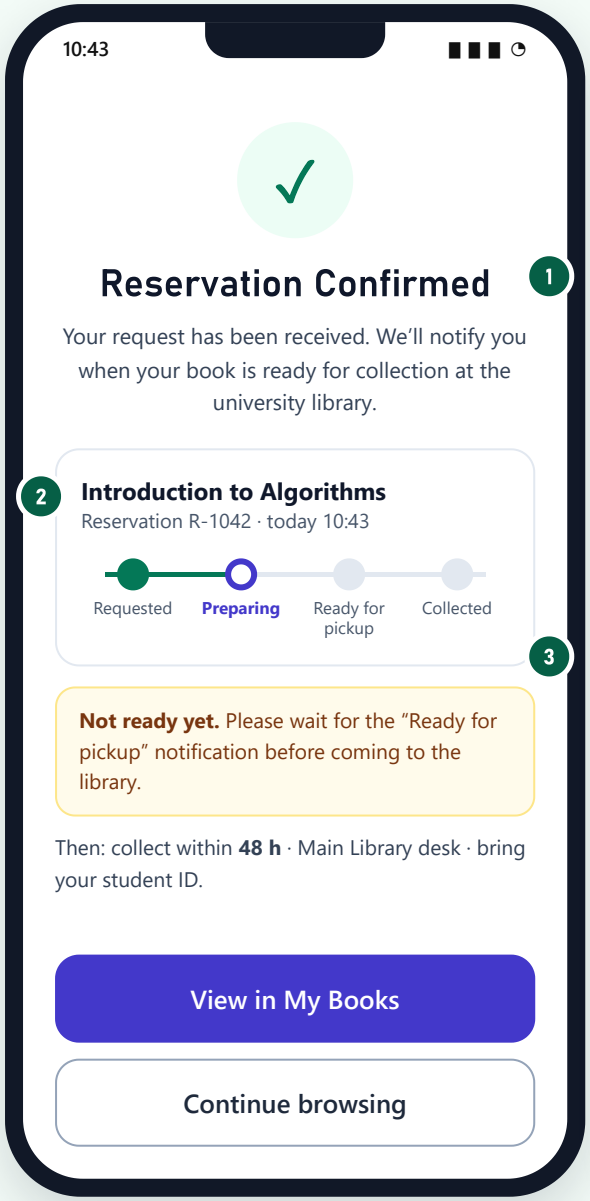
1. **Why** sign-in is needed is explained; the chosen book is shown in the sheet.
2. **University identity,** not a custom form — no ID-number-only login, no name/faculty fields FR-05 .
3. **Labels above inputs,** show-password, recovery link, large targets.
4. **Eligibility** confirmed from the university record FR-06 ; nothing re-typed.
5. **Returns to the task** with the book preserved FR-08 and sets expectations before confirming.

**If sign-in fails:** “We couldn’t sign you in. Check your username and password, or reset it.” The reservation is not lost; the user returns to C4·a.

PART C

# Confirmation → My Books

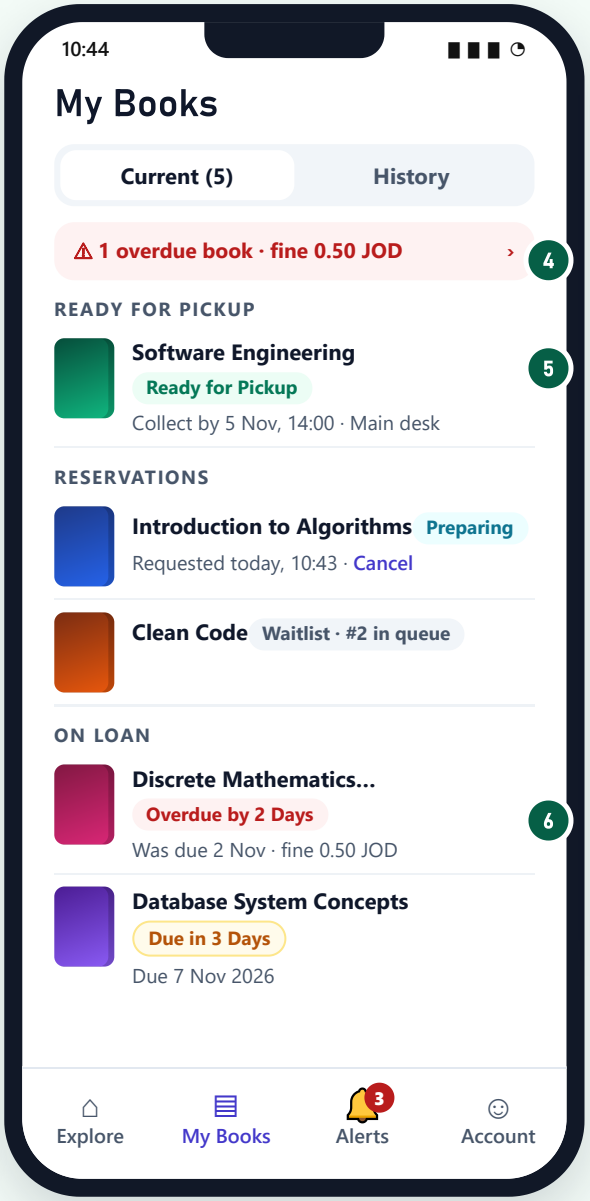
Transparent status at every stage FR-12 FR-13 FR-25–26




C5

**Reservation confirmed**

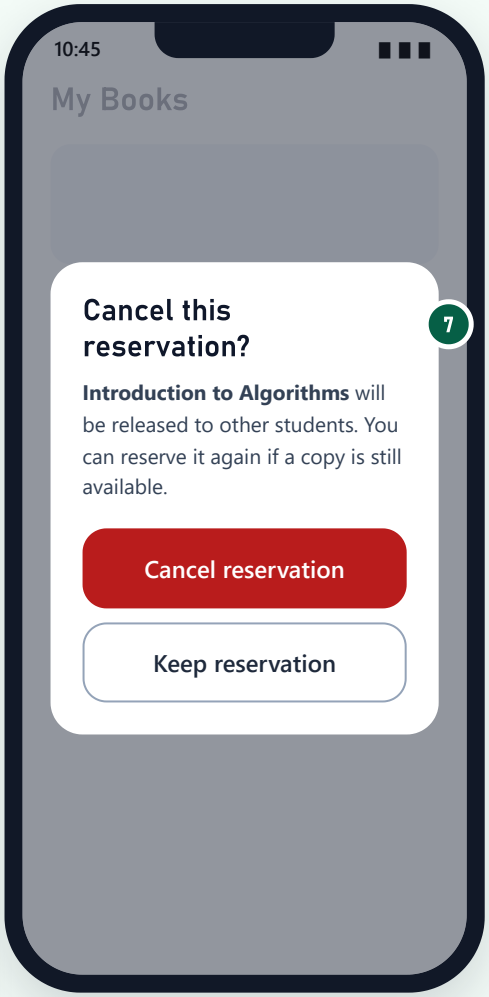
Honest: received, not ready




C6

**My Books**

Grouped by what Omar must do




C6

**Confirm destructive action**

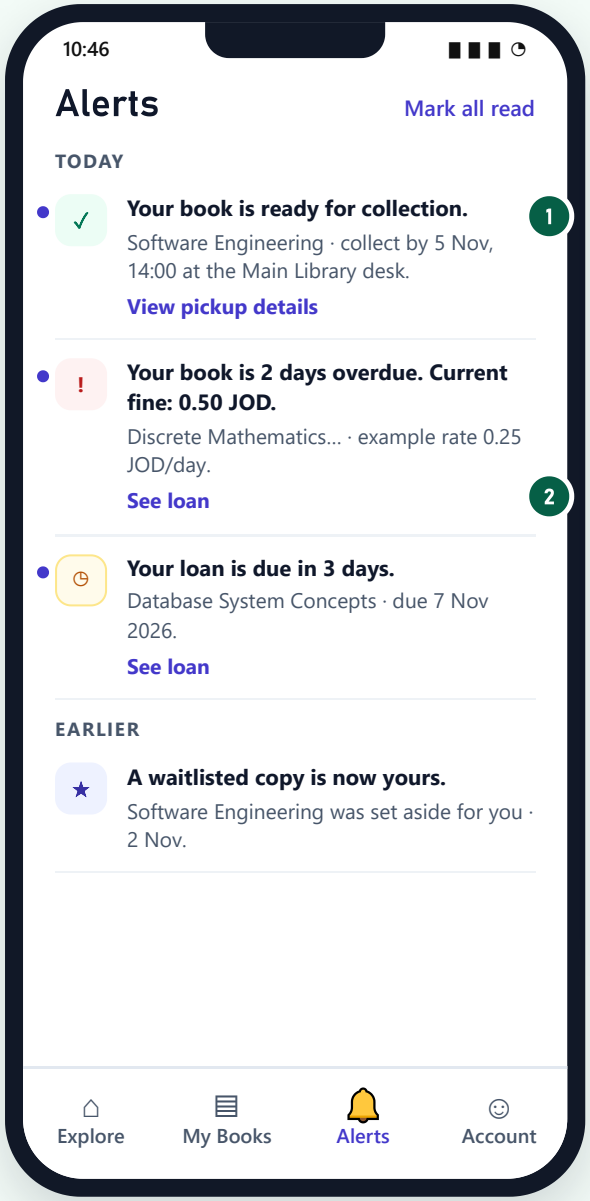
After tapping "Cancel"

1. **Clear success** that doesn't claim the book is ready.
2. **Progress tracker** shows where the request is in the process.
3. **Physical next steps** stated up-front; shortcut to My Books.
4. **Problems first:** overdue banner with the fine and currency.
5. **Grouped by action** — collect, wait, return. Reservation ≠ ready.
6. **Plain status labels** and absolute dates, no codes.
7. **Destructive action confirmed,** consequence explained, safe option available.

PART C

# Notifications, loading and error states

Useful messages with actions FR-14–18 · honest failure E8




C7

**Notifications**

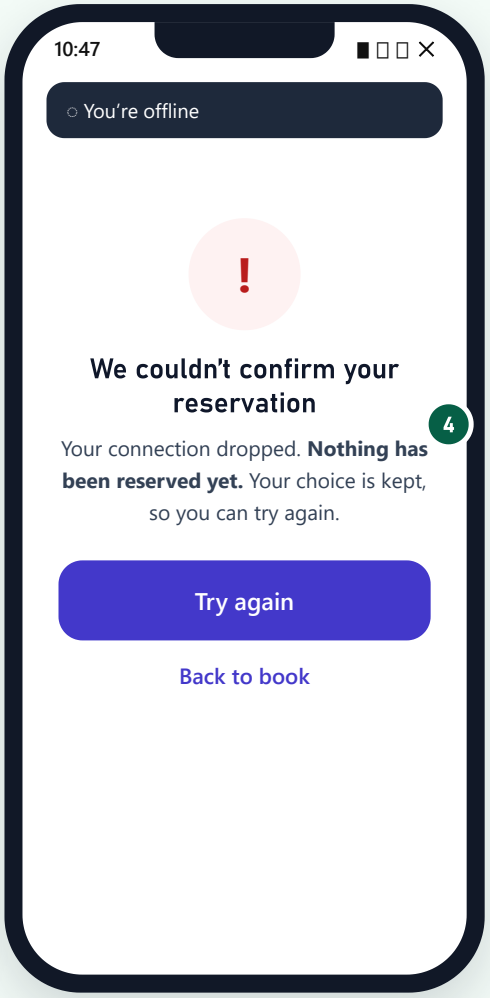
Each one has a next action




State

**Loading**

Skeleton keeps the layout stable




State

**Error**

Lost connection while confirming

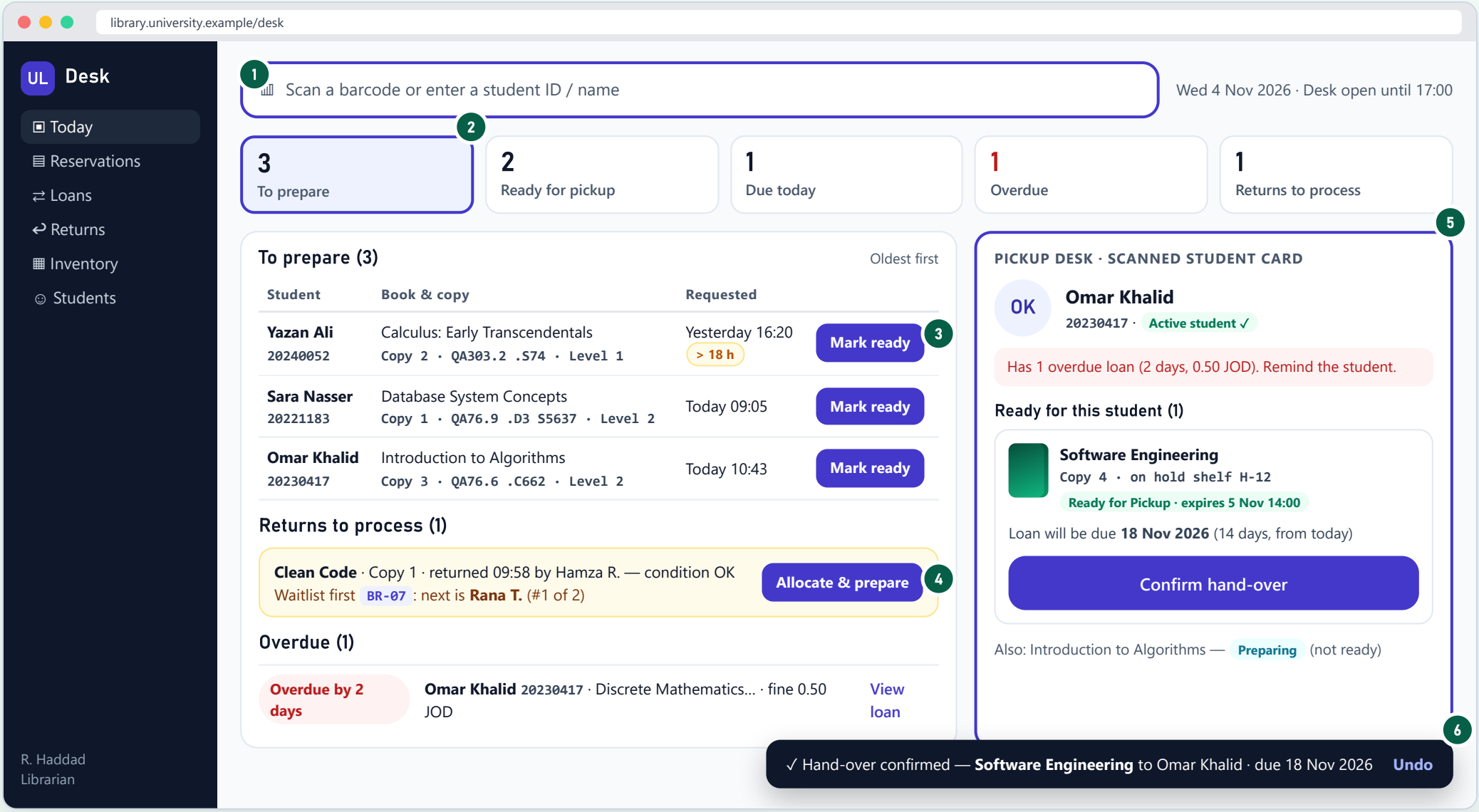
1. **Plain-language messages** matching the SRS examples; unread items marked with a dot and bold text, not colour alone.
2. **Fines are transparent:** days overdue, amount, currency and the (illustrative) rate FR-17 BR-03 .
3. **Loading:** skeleton cards plus text, so screen-reader users also know something is happening.
4. **Error:** says what happened, what did *not* happen, and how to recover. Never a false success.

Reminders follow FR-16 : 3, 2 and 1 day before, and on the due date — also sent to university email per the communication policy.

PART C

# Librarian dashboard

Desktop-first: used all day at a desk with a barcode scanner FR-19-24




1. **Scan-first search** for student or copy.
2. **Task queues,** not one table of codes.
3. **One action per row,** labelled with the verb.
4. **Waitlist rule built in** to returns.
5. **Student identity** and eligibility first.
6. **Specific feedback + Undo.**

PART C

# Visual system & accessibility decisions

One consistent set of rules across student and librarian screens

## Colour roles & contrast on white

**Primary · #4338CA**
buttons, links, focus · 7.9 : 1

**Success · #047857**
available, ready · 5.5 : 1

**Warning · #B45309**
due soon, waitlist · 5.0 : 1

**Danger · #B91C1C**
overdue, destructive · 6.5 : 1

**Secondary text · #475569**
metadata · 7.6 : 1

All exceed WCAG 2.2 AA (4.5 : 1 for text). Status is never shown by colour alone — always text + icon.

## Typography (mock-ups at ~73% scale)

**Screen title**

**Section heading**

Body text — 16 px on device

Metadata — 13–14 px on device

Absolute dates ("7 Nov 2026") plus relative labels ("Due in 3 days"). No status codes or abbreviations.

## Touch & focus

Primary buttons 48 px tall on device; all targets ≥ 44 px (above the WCAG 2.2 minimum of 24 px). Visible focus ring for keyboard users on the librarian dashboard.

Focused focus ring

## Components

Primary — Reserve book

Secondary — Continue browsing

Destructive — Cancel reservation

Disabled — with reason shown

## Status labels

• 2 of 3 available | Preparing | Ready for Pickup | Due in 3 Days | Overdue by 2 Days | Waitlist · #2 | Returned

The same labels appear in the student app, notifications and librarian dashboard.

**Labels** sit above inputs, never placeholder-only.

**Errors** appear next to the field, in words, and keep what the user typed.

**Navigation** — 4 labelled tabs, identical on every student screen.

**Screen readers** — status badges and icons have text equivalents.

PART D

# Design comparison & answer key (1/2)

Student-facing screens A1–A3

| # | Screen | Problem in bad design | Improvement in good design | UX principle | Type |
|---|---|---|---|---|---|
| 1 | A1 | **Login required before browsing;** search hidden until sign-in. Contradicts `FR-07`. | Browse and search as a guest; authenticate only at "Reserve" (C1, C4). | Reduce friction · user goals first | Requirements, Flow |
| 2 | A1 | **Custom form asks for ID, password, name, faculty, year, email** — data the university already has; not an approved identity mechanism. | University single sign-on; profile and eligibility come from university records (C4·b). | Minimise input · security by design | Requirements, Interaction |
| 3 | A1 | **Weak hierarchy:** giant institutional header, two notices, welcome paragraph and news compete with the task. | Search is the largest element; one short "How borrowing works" card. | Visual hierarchy · minimalist design | Visual |
| 4 | A1 | **Cramped, cryptic navigation** ("Svc", "E-Res", "OPAC", "MyAcc") with tiny targets. | 4 labelled bottom tabs: Explore, My Books, Alerts, Account. | Match the real world · recognition | IA, Accessibility |
| 5 | A1 | **Tiny checkbox and links;** "Reset" next to and styled like "Submit Form". | ≥ 44 px targets; one primary button; no reset button. | Error prevention · Fitts's law | Accessibility, Interaction |
| 6 | A1 | **Low-contrast grey text** (≈ 2.5 : 1) and "best viewed at 1280×1024" — not designed for phones. | All text ≥ 4.5 : 1; mobile-first layout. | WCAG 1.4.3 · mobile-first | Accessibility, Visual |
| 7 | A2 | **Overloaded results:** tiny covers, publisher, pages, call number, location codes on every row. | Cover, title, author, edition, availability; details moved to the book page. | Information hierarchy · progressive disclosure | Visual, IA |
| 8 | A2 | **Ambiguous availability:** "In (2/3)", "Out", "Avl", "Lib Use Only" with a green dot; colour-only meaning; footnote says status may be wrong. | "2 of 3 copies available" / "All copies on loan · 1 waiting" with icon + text. | Clarity · visibility of status · WCAG 1.4.1 | IA, Accessibility |
| 9 | A2 | **Four identical buttons** per row; "Hold" vs "Request" meaning unclear; both offered for unavailable books. | Whole card is tappable; one context-correct action on the detail page. | Consistency · recognition over recall | Interaction |
| 10 | A2 | **Filters hidden** in a tiny "Advanced options" link after pagination; weak search box. | Prominent search with clear button; filter chips including "Available now". | Discoverability · efficiency | IA |
| 11 | A3 | **Wrong priority:** "Print Record" and "Export (MARC)" are prominent; reservation buttons are faint and at the bottom. | Sticky "Reserve book" as the only primary action. | Primary action clarity | Visual, Interaction |
| 12 | A3 | **Librarian jargon:** MARC fields, "Holdings: 3 \| Status: see shelf" — not whether a copy can be reserved. | Plain metadata, collapsed "Details"; availability card first. | Match the real world | IA |
| 13 | A3 | **Hidden physical requirements:** pickup inside a collapsed "Notice"; loan period in 6.8 px footer. | Pickup place, 14-day loan and 48-hour hold shown as rows before reserving. | Set expectations · visibility | IA, Requirements |

PART D

# Design comparison & answer key (2/2)

Screens A4–A7

| # | Screen | Problem in bad design | Improvement in good design | UX principle | Type |
|---|---|---|---|---|---|
| 14 | A4 | **Repeated data entry:** ID, name, email and ISBN typed again although the user is signed in and on the book page. | Nothing re-typed; review screen shows the book and identity (C4·c). | Efficiency · minimise memory load | Interaction |
| 15 | A4 | **Vague, irrelevant fields:** "Request Type: Normal / ILL / Recall", "Ref.", a one-option "Pickup Point", "Required by". | Only what the SRS needs; pickup location shown as information. | Simplicity · requirements traceability | Requirements IA |
| 16 | A4 | **No progress indication;** "Next >>" leads to more steps; Cancel is styled as the primary button. | Single review + confirm; primary button is the forward action. | Visibility of status · user control | Flow Visual |
| 17 | A4 | **Unhelpful error** ("Error 0x1F") that clears every field; placeholder-only date format; 5-minute timeout. | Errors next to the field in plain words; input kept; no free-text date needed. | Help users recover · error prevention | Interaction Accessibility |
| 18 | A1→A4 | **After login the user lands on Home,** not the book they chose — they must search again. Contradicts `FR-08`. | Return to the same reservation after sign-in. | Preserve context | Requirements Flow |
| 19 | A5 | **Misleading success:** "processed successfully" suggests the book is ready; status "PND-Q"; collection rules in tiny grey text far below. | "Reservation Confirmed … we'll notify you when ready"; progress tracker; "Not ready yet" notice; pickup steps. | Honest feedback · set expectations | IA Visual |
| 20 | A5 | **Dead end:** only Print / New Search / Home — no route to track the reservation. | "View in My Books" primary; "Continue browsing" secondary. | Navigation · next steps | Flow |
| 21 | A6 | **One table of codes** (R/W/L, PND/RDY/ACT/OD/RTN); "Ready" looks the same as "Pending"; overdue not highlighted. | Grouped by what to do: Ready / Reservations / On loan; plain labels and colours + text. | Visibility of status · recognition | IA Visual |
| 22 | A6 | **Mixed date formats:** "11/07/26" (US order) next to "24/10/26" and ISO dates; "Due" column reused for pickup expiry. | "Due 7 Nov 2026 · Due in 3 Days"; pickup deadline labelled "Collect by". | Consistency · clarity | Visual IA |
| 23 | A6 | **Fine "0.500"** without currency, rate or reason; "Reg. 9.4" reference only. | "Overdue by 2 Days · fine 0.50 JOD (0.25 JOD/day, illustrative)". | Trust · transparency | IA |
| 24 | A6 | **Notifications hidden** as "Messages (3)" text link; "Cancel Selected" acts without confirmation. | Alerts tab with badge; actionable messages; confirmation dialog for cancel. | Visibility · error prevention | Interaction IA |
| 25 | A7 | **15-column table** where session IDs and timestamps outshine student ID (in light grey); all statuses look alike. | Task queues (To prepare, Ready, Overdue, Returns); student name + ID prominent. | Workflow optimisation · hierarchy | Visual IA |
| 26 | A7 | **Slow actions:** select row → dropdown → "Go"; exact-ID filter; feedback "1 record(s) affected". | Scan-first search; one labelled button per row; specific toast with Undo. | Efficiency · feedback · user control | Interaction |
| 27 | A7 | **Waitlist rule not visible** at check-in; the librarian could reshelve a copy someone is waiting for. | Returns card shows "Waitlist first: next is Rana T." with one action `BR-07`. | Error prevention · encode business rules | Requirements Flow |

**Type:** Visual visual design · Interaction interaction design · IA information architecture · Accessibility · Flow user flow · Requirements requirements misunderstanding

PART D

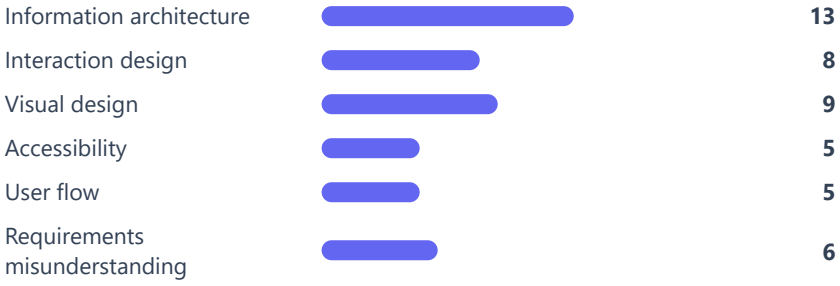
# Key differences & how to use this reference

| Problem in bad design | Improvement in good design | UX principle |
|---|---|---|
| Login required before browsing | Browse first, authenticate at reservation | Reduce friction |
| Cluttered catalogue | Clear search and meaningful filters | Information hierarchy |
| Ambiguous availability | Availability per physical copy | Clarity and feedback |
| Confusing reservation steps | Clear guided reservation flow | Efficiency |
| Hidden physical collection requirement | Explicit pickup instructions | Set expectations |
| Ambiguous loan statuses | Clear status labels | Visibility of system status |
| Confusing fine information | Transparent overdue information | Trust and transparency |
| Difficult librarian operations | Task-focused dashboard | Workflow optimisation |

### Bad UX vs. missing functionality

The Part A design **can do everything** the SRS asks — browse, search, reserve, waitlist ("Hold"), cancel, notifications ("Messages"), fines, and every librarian action (via the "Action" menu). Most problems are therefore **UX problems**: the capability exists but is slow, hidden or misleading. Three items are different: rows **1**, **2** and **18** directly contradict requirements ( `FR-07` , `FR-05` , `FR-08` ) — these are **requirements misunderstandings**, not matters of taste.

### Issues by type (27 total)




Many issues have more than one type, so counts overlap.

### This is a proposed reference, not the answer

Several choices are trade-offs: e.g. hiding bibliographic detail helps students but may slow researchers; a 4-tab bar is simple but leaves no room for future features. Before building, a real team would **validate it with usability testing**:

- 5–8 students: find a book, reserve it, find what to collect and what is owed.
- 2–3 librarians: process a morning queue and a busy pickup hour.
- Measure task success, time on task, errors and confidence; then iterate.

Designing software means understanding **the users**, **their goals**, **the requirements** and **the consequences of every design decision**.